

THE AI ENGINEER'S INTERVIEW PREP KIT

100

AI INTERVIEW QUESTIONS

Curated for 2026 hiring loops · LLMs · RAG · MCP · AI Agents · Agentic AI



Question distribution across the 9 sections — 80% focused on the modern AI stack

07

Math, Probability & Statistics

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Classical Machine Learning

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Deep Learning & Neural Networks

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Large Language Models

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Retrieval-Augmented Generation

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Evaluation & Metrics

08

Model Context Protocol (MCP)

12

AI Agents

13

Agentic AI

Every question here is the kind that actually gets asked — no obscure theory, no filler.

01

Math, Probability & Statistics

The foundations every interviewer loves to warm up with

Q1–Q7
7 questions

1

Explain the bias–variance tradeoff. How does it show up as underfitting vs. overfitting, and how do you manage it in practice? **HIGH-YIELD**

2

State Bayes' theorem and walk through a practical example (e.g., spam filtering or medical test false positives). **HIGH-YIELD**

3

What is the difference between probability and likelihood?

4

What is a p-value? What does $p < 0.05$ actually mean — and what does it NOT mean?

5

Explain the Central Limit Theorem and why it is so important in statistics and ML.

6

What are Type I and Type II errors? Give a real-world example where each one is more costly.

7

Differentiate correlation, covariance, and causation. Why doesn't correlation imply causation?

02

Classical Machine Learning

Bread-and-butter ML that still shows up in every round

Q8–Q14
7 questions

8

Explain supervised, unsupervised, and reinforcement learning with one real example of each.

9

How does logistic regression differ from linear regression? Why can't we use plain linear regression for classification? **HIGH-YIELD**

10

How do you detect overfitting, and what techniques do you use to prevent it (regularization, cross-validation, early stopping)? **HIGH-YIELD**

11

Compare Random Forest and Gradient Boosting (e.g., XGBoost). How does bagging differ from boosting? **HIGH-YIELD**

12

How does k-means clustering work? How do you choose the right value of k?

13

Your dataset is highly imbalanced (e.g., 1% fraud). How do you handle it — resampling, class weights, and which metrics do you use? **HIGH-YIELD**

14

What is the difference between L1 (Lasso) and L2 (Ridge) regularization? Why does L1 produce sparse models?

03

Deep Learning & Neural Networks

Core neural-net concepts, asked almost verbatim

Q15–Q20
6 questions

- 15 Explain backpropagation intuitively. What is actually being computed and updated? **HIGH-YIELD**
- 16 What are vanishing and exploding gradients? What causes them and how do we fix them (ReLU, residual connections, normalization, clipping)? **HIGH-YIELD**
- 17 Compare activation functions: sigmoid, tanh, ReLU, GELU. Why is ReLU the common default?
- 18 What is the difference between Batch Normalization and Layer Normalization? Why do Transformers use LayerNorm?
- 19 How does dropout work? Why does it behave differently at training time vs. inference time?
- 20 When would you use a CNN vs. an RNN vs. a Transformer? What limitations of RNNs did Transformers solve? **HIGH-YIELD**

04

Large Language Models

Transformers, parameters, fine-tuning & inference — the heart of modern interviews

Q21–Q40
20 questions

- 21 Explain the Transformer architecture at a high level. What made 'Attention Is All You Need' such a breakthrough? **HIGH-YIELD**
- 22 Explain self-attention. What are Query, Key, and Value, and how is the attention score computed (scaled dot-product)? **HIGH-YIELD**
- 23 Why do Transformers use multi-head attention instead of a single attention head?
- 24 Why do Transformers need positional encodings? What is RoPE (rotary positional embedding)?
- 25 Compare encoder-only (BERT), decoder-only (GPT), and encoder-decoder (T5) architectures. When is each used? **HIGH-YIELD**
- 26 What is tokenization? Explain Byte-Pair Encoding (BPE) and how token count affects cost and context limits. **HIGH-YIELD**
- 27 Explain temperature, top-p (nucleus), and top-k sampling. When would you set temperature to 0? **HIGH-YIELD**

28

What is a context window? What happens when it is exceeded, and what is the 'lost in the middle' problem? **HIGH-YIELD**

29

Differentiate pre-training, supervised fine-tuning (SFT), and instruction tuning.

30

Explain RLHF end-to-end: the role of the reward model and policy optimization. How is DPO a simpler alternative? **HIGH-YIELD**

31

What is parameter-efficient fine-tuning? Explain LoRA and QLoRA and why they are so widely used. **HIGH-YIELD**

32

What is model quantization (8-bit, 4-bit)? What tradeoffs does it introduce?

33

Why do LLMs hallucinate? List practical mitigation strategies you would use in production. **HIGH-YIELD**

34

Compare zero-shot, few-shot, and chain-of-thought prompting. When does CoT actually help? **HIGH-YIELD**

35

What is the KV cache and why is it critical for fast autoregressive inference?

36

Explain common inference parameters: max tokens, stop sequences, frequency penalty, and presence penalty.

37

Prompt engineering vs. RAG vs. fine-tuning: how do you decide which approach a use case needs? **HIGH-YIELD**

38

What is Mixture of Experts (MoE)? How does it let models scale parameters without scaling inference cost proportionally?

39

What are prompt injection and jailbreaking? How would you defend an LLM application against them? **HIGH-YIELD**

40

What is the difference between a system prompt and a user prompt? How do structured outputs / JSON mode work?

05

Retrieval-Augmented Generation

Pipelines, embeddings, chunking, vector DBs & advanced RAG

Q41–Q55
15 questions

41

What is Retrieval-Augmented Generation? What problems does it solve compared to fine-tuning or relying on model memory? **HIGH-YIELD**

42

Walk through a complete RAG pipeline end-to-end: ingestion, chunking, embedding, indexing, retrieval, and generation. **HIGH-YIELD**

43

What are embeddings? How is similarity between two embeddings measured, and why is cosine similarity common? **HIGH-YIELD**

44

Explain chunking strategies (fixed-size, recursive, semantic). How do chunk size and overlap affect retrieval quality? **HIGH-YIELD**

45

What is a vector database? Name a few (FAISS, Pinecone, Chroma, pgvector) and explain how ANN indexes like HNSW enable fast search. **HIGH-YIELD**

46

Compare dense retrieval and sparse/keyword retrieval (BM25). What is hybrid search and when is it worth it? **HIGH-YIELD**

47

What is reranking and why add it after retrieval? Explain bi-encoder vs. cross-encoder.

48

How should a RAG system behave when retrieval returns nothing relevant? How do you prevent it from answering anyway?

49

Explain query transformation techniques: query rewriting, multi-query, and HyDE.

50

Describe advanced RAG patterns you know: parent-document retrieval, sentence-window retrieval, and GraphRAG.

51

How do you keep the vector index in sync when source documents are added, updated, or deleted?

52

How do you evaluate a RAG system? Explain faithfulness, answer relevance, and context precision/recall (e.g., RAGAS). **HIGH-YIELD**

53

What are the most common RAG failure modes, and how do you debug whether the problem is retrieval or generation? **HIGH-YIELD**

54

How do you implement metadata filtering and user-level access control in a RAG system?

55

With million-token context windows, do we still need RAG? Argue both sides. **HIGH-YIELD**

06

Evaluation & Metrics

How to prove your model or system actually works

Q56–Q67
12 questions

56

Define precision, recall, and F1 score. Give a scenario where you'd optimize for precision and one for recall. **HIGH-YIELD**

57

When is accuracy a misleading metric? Explain ROC-AUC vs. PR-AUC and when to prefer each. **HIGH-YIELD**

58

Compare regression metrics: MSE, RMSE, MAE, and R^2 . When would you prefer MAE over RMSE?

59

What is perplexity and what does it measure for a language model? What are its limitations?

HIGH-YIELD

60

Explain BLEU and ROUGE. What do they measure and why are they poor fits for open-ended LLM outputs?

HIGH-YIELD

61

What is BERTScore / semantic-similarity-based evaluation, and how does it improve on n-gram metrics?

62

Explain LLM-as-a-judge. What biases (position bias, verbosity bias, self-preference) must you control for?

HIGH-YIELD

63

What do popular benchmarks measure — MMLU, HumanEval, GSM8K, MT-Bench? Why can benchmark scores be misleading?

64

How is human evaluation done at scale? Explain pairwise comparisons and Elo-style leaderboards like Chatbot Arena.

65

How would you evaluate a RAG system end-to-end, separating retrieval quality from generation quality?

HIGH-YIELD

66

How do you evaluate an AI agent — task success rate, tool-call accuracy, and trajectory/step-level evaluation?

HIGH-YIELD

67

How do you set up evaluation for an LLM product in production: golden datasets, regression tests, A/B testing, and drift monitoring?

HIGH-YIELD

07

Model Context Protocol (MCP)

The new standard for connecting LLMs to tools & data

Q68–Q75
8 questions

68

What is the Model Context Protocol (MCP), and what 'M×N integration problem' does it solve?

HIGH-YIELD

69

Explain the MCP architecture: host, client, and server. What role does each play?

HIGH-YIELD

70

What are MCP's core primitives — tools, resources, and prompts? How does each get used by the model or the user?

HIGH-YIELD

71

What transports does MCP support (stdio, streamable HTTP/SSE)? When would you use local stdio vs. a remote server?

- 72 How is MCP different from plain function calling or a custom plugin/API integration? **HIGH-YIELD**
- 73 Walk through an end-to-end MCP tool call: discovery, invocation, and how the result gets back into the model's context.
- 74 What are the security risks with MCP — tool poisoning, prompt injection through tool results, credential exposure — and how do you mitigate them? **HIGH-YIELD**
- 75 When would you build an MCP server for your product vs. just exposing a REST API? What do you gain?

08

AI Agents

Tools, memory, planning loops & agent engineering

Q76–Q87
12 questions

- 76 What is an AI agent? How is it different from a plain LLM chatbot or a fixed workflow? **HIGH-YIELD**
- 77 What are the core components of an agent — the LLM 'brain', tools, memory, and the reasoning/acting loop? **HIGH-YIELD**
- 78 Explain the ReAct pattern. Why does interleaving reasoning and action beat one-shot generation for complex tasks? **HIGH-YIELD**
- 79 How does function/tool calling work under the hood? How does the model decide which tool to call and who executes it? **HIGH-YIELD**
- 80 Explain short-term vs. long-term memory for agents. How would you implement persistent memory (e.g., with a vector store)? **HIGH-YIELD**
- 81 How do agents plan? Compare plan-and-execute with iterative ReAct-style task decomposition.
- 82 How do you handle agent failure modes: infinite loops, repeated tool errors, and runaway costs (max iterations, timeouts, retries)? **HIGH-YIELD**
- 83 Compare agent frameworks — LangGraph, CrewAI, AutoGen. When would you skip frameworks and build directly on the model API? **HIGH-YIELD**
- 84 Where would you insert human-in-the-loop approval in an agent, and why is it essential for high-stakes actions?
- 85 What guardrails would you put around an agent — input/output validation, sandboxed execution, and scoped permissions? **HIGH-YIELD**

86 How do you safely give an agent access to private or sensitive data?

87 How do you make agents observable in production — tracing, logging every tool call, and replaying trajectories?

09

Agentic AI

Multi-agent systems, orchestration & production autonomy

Q88–Q100
13 questions

88 What does 'agentic AI' mean? Describe the spectrum from a single LLM call to a fully autonomous system. **HIGH-YIELD**

89 Workflows vs. agents: explain patterns like prompt chaining, routing, parallelization, orchestrator–workers, and evaluator–optimizer. When is a workflow enough? **HIGH-YIELD**

90 Describe multi-agent architectures: supervisor, hierarchical, and peer-to-peer/swarm. Give a use case for each. **HIGH-YIELD**

91 How do agents in a multi-agent system communicate and coordinate — shared state vs. message passing? What is the A2A idea?

92 Single powerful agent vs. multiple specialized agents: what are the cost, latency, and complexity tradeoffs? **HIGH-YIELD**

93 How do you manage state and orchestration in a long-running agentic workflow (checkpointing, resumability, durable execution)?

94 What is context engineering for agents? How do you handle context-window limits over long tasks (summarization, compaction, scratchpads)? **HIGH-YIELD**

95 How do you optimize cost and latency of an agentic system — model routing (small vs. large models), prompt caching, and parallel tool calls? **HIGH-YIELD**

96 Errors compound across multi-step trajectories: if each step is 95% reliable, a 20-step task mostly fails. How do you engineer reliability into agentic pipelines? **HIGH-YIELD**

97 What safety concerns are unique to agentic AI — irreversible actions, permission scoping, sandboxing — and how do you address them? **HIGH-YIELD**

98 How do you evaluate agentic systems — task completion rate, trajectory analysis, and benchmarks like SWE-bench or GAIA?

99 Pick a real agentic use case (coding agent, deep research, customer support automation). What makes it hard in practice?

100

What are the current limitations of agentic AI (long-horizon planning, reliability, cost), and where do you see it going in the next 2 years?

HIGH-YIELD